

Do Markets Price the Macroeconomic Content of AI Events?

High-frequency AI-exposed earnings and infrastructure shocks

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What This Project Shows

- Builds a source-audited AI-event database for earnings, infrastructure, and frontier-model news.
- Converts event-window market reactions into a monthly AI-exposed event-shock proxy.
- Estimates reduced-form local projections, Anderson–Rubin weak-IV sets, conditional LP-IV, and firm-exposure regressions.
- Separates what is strongly supported from what is conditional on a borderline first stage.

The paper is the technical record. This deck is a fast reading path for an interviewer.

One-Slide Answer

- The current event-shock proxy has complete selected-event coverage: 135/135 finite shocks.
- Reduced-form LPs are the cleanest result; equity-market outcomes and firm-exposure patterns carry the strongest signal.
- The preferred LP-IV first stage is borderline: robust $F = 14.994$, partial $R^2 = 0.1394$.
- Anderson–Rubin sets therefore lead the causal discussion.
- The 2017–2022 extension passes coverage but fails relevance: 212/212 finite shocks, $F = 4.611$.

Identification Hierarchy

Object	Role in the paper
Reduced-form LP	Main evidence: responses to observed event-window repricing.
Anderson–Rubin sets	Weak-IV robust causal uncertainty statement.
Conditional LP-IV	Secondary relative impulse responses normalized by n_t .
BVAR, sign restrictions, state-space	Interpretation and robustness, not a substitute for instrument relevance.

The project does not claim a fully observed structural AI-capex shock.

Current sample

- 135 AI-exposed events.
- 2022–2026 baseline.
- LSEG/Refinitiv timestamps where available.
- Confound calendar for macro, earnings, and unscheduled news.

Timing rule

- Use the earliest audited market-observable headline or earnings-release timestamp.
- Map off-hours events to the next regular session.
- Keep caveated timestamps only with timing/drop sensitivity.

Shock Construction

Event-window abnormal AI-basket return

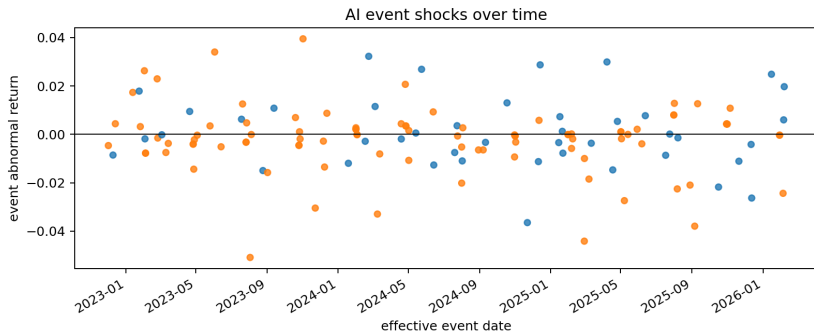
$$r_{B,e}^{(w)} = \sum_{j \in B_e} \omega_{j,e} r_{j,e}^{(w)}, \quad z_e^{(w)} = r_{B,e}^{(w)} - \hat{\beta}_B r_{M,e}^{(w)}.$$

Monthly proxy

$$z_t = \sum_{e \in t} z_e^{(w)}.$$

- Baseline: 60-minute post-event window, QQQ adjustment, event-day-excluded beta.
- SPY, narrower windows, event drops, and stronger-shock screens are robustness checks.

Event Timeline



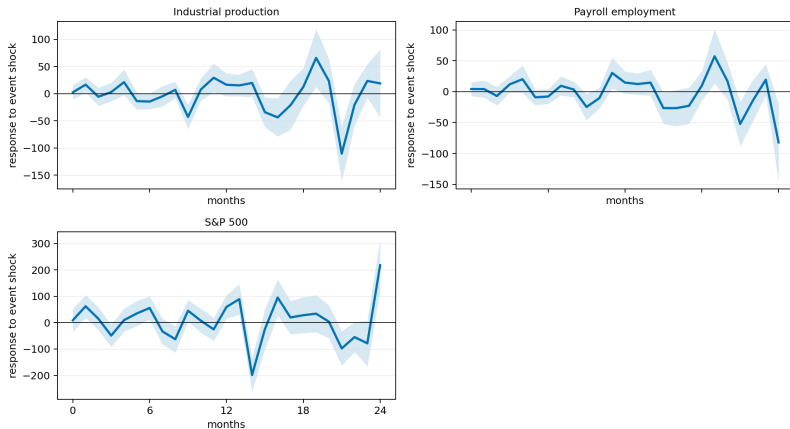
Lag-augmented local projection

$$y_{t+h} - y_{t-1} = \alpha_h + \theta_h z_t + \Gamma_h X_{t-1}^{(\rho)} + u_{t+h}.$$

- θ_h is the response to the event-shock proxy.
- Lag augmentation addresses persistence and horizon-specific inference concerns.
- HC1 Eicker–Huber–White intervals lead; HAC/bootstrap checks stay in the appendix.

Reduced-Form Responses

Reduced-form event-shock responses



Reduced-form responses are the headline object because the first stage is borderline.

Why LP-IV Is Secondary

- The preferred first stage is informative but not strong enough for a clean structural claim.
- Robust $F = 14.994$ is below the strong-IV threshold used in the paper.
- Anderson–Rubin intervals are available for 175 rows and lead the causal uncertainty discussion.
- Conditional LP-IV point estimates are still useful, but only after the reduced-form evidence.

Stronger-Shock Screens

Candidate	Best robust F	Treatment
Capex/infrastructure-SPY	20.443	Comparator
Capex/cloud guidance text	1.221	Diagnostic
Structured guidance surprise	3.464	Diagnostic
Expectation-surprise import	3.485	Diagnostic

The candidate screens improve measurement discipline but do not overturn the reduced-form-first hierarchy.

Firm-Level Exposure Evidence

- A broad CRSP/Compustat/ESZ/IBES panel tests whether AI-exposed event quarters map into firm outcomes.
- The firm-level table is useful as reduced-form heterogeneity evidence.
- Interaction-IV columns remain visible but are labeled as weak-IV companion estimates.
- This is a second evidence pillar, not a separate strong causal design.

Tooling and Reproducibility

- Python pipeline for ingestion, event construction, shock construction, LP/LP-IV estimation, robustness, and paper outputs.
- Data interfaces for WRDS, LSEG/Refinitiv, Databento, FRED, SEC, BEA, EIA, and Polygon.
- Tests cover data-acquisition interfaces, shock construction, first stages, LP-IV, expectation-surprise screens, timing sensitivity, and paper artifacts.
- Public repo keeps code and artifacts, not raw licensed data or credentials.

Takeaway

The project is strongest as a disciplined market-reaction design.

- AI events contain measurable macro-financial information.
- The data support reduced-form responses and weak-IV robust uncertainty.
- Stronger structural AI-capex claims require broader pre-event capex/cloud expectation data.

Backup: Current Empirical Gate

- Validity gate: borderline-IV; AR intervals lead.
- Selected shock:
`post_60m__QQQ__event_excluded_full__all_events__z_eq.`
- Current-sample intraday gaps: 0.
- Baseline status: The empirical design uses a complete current-sample event-shock proxy, but validity and relevance diagnostics require reduced-form responses and Anderson-Rubin sets to lead the causal interpretation.

Backup: Extension Sample

- Extension events: 212.
- Finite extension shocks: 212.
- Preferred Tier 2 timestamps: 6.
- Public/newswire fallback Tier 2 timestamps are appendix-only: 14.
- First-stage relevance remains weak: $F = 4.611$.

Backup: What Would Strengthen the Paper

- Timestamped StreetAccount or equivalent capex/cloud consensus.
- Broader pre-event segment revenue and capex expectations.
- Independent benchmarks for transcript-coded surprises.
- A first stage that survives event-window-excluded normalization.